

Brijodison

Engineering Notebook Element F

Engineering Service Learning

Consideration of Design Viability

We believe that our proposed design was critically evaluated using extra-functional considerations. We had many items that we looked at to make sure that our product would work well. While working on our project we took into consideration how realistic and connected to the solution we could make our product, if we supported our product right, and if our solution stayed problem focused.

For taking into account the realisticness we made sure to talk to Mr.Kiser and ask him valuable questions that could help us better our creation. We asked certain questions like what part of his back hurts the most when he pushes the cart or how we can make the handle of the cart feel better. Once we had gotten that information we looked back to our design matrix and requirements to see how we could combine what we learned to make an effective product. We believe that the solution that we have come up with satisfies this requirement of making sure it was realistic in context of our prompt.

While working we checked to make sure that we supported our product right. When prototyping we made sure to take all evidence we acquired into account when creating this product. It was very important for us to not only get the right evidence but to also apply right to our solution. We made sure to listen to the advice that our stakeholder, Mr. Kiser, gave us. We had Mr. Kiser test our prototype to see if it was right for him and where we could improve. We also wanted to make sure that we could effectively use our evidence to build a successful

prototype. We looked at all of our recent and past evidence to ensure that our design fit the requirements and that we stayed true to the realisticness.

One of the last important pieces that we made sure to take into account was did our solution stay problem focused. For most engineering problems it gets easy to lose sight of the actual problem that we are trying to solve. Some might get carried away and forget that they aren't just trying to come up with solutions but they are trying to help someone else. In our case what we were trying to solve or better was the pain that Mr. Kiser felt in his lower back when pushing the cart. Since Mr. Kiser is taller than most people that push these types of carts, it is uncomfortable for him to bend down and push the cart. We were tasked with finding a solution that would better the pain he experiences. We made sure that we took all of this information into consideration while prototyping. While also taking in the evidence he gave us we also looked at the certain requirements we had to meet to make sure that it was an effective product. We wanted to be creative but still kept in mind that the main person we were trying to benefit and help was Mr. Kiser.

When coming to the end of designing and making our prototype we looked back to the requirements and made sure that this was an effective solution to the problem. We made sure that our product was realistic, well-supported, and problem-focused. We wanted to create a prototype that was successful but also followed the right engineering steps to make it beneficial to the people that we are trying to help.